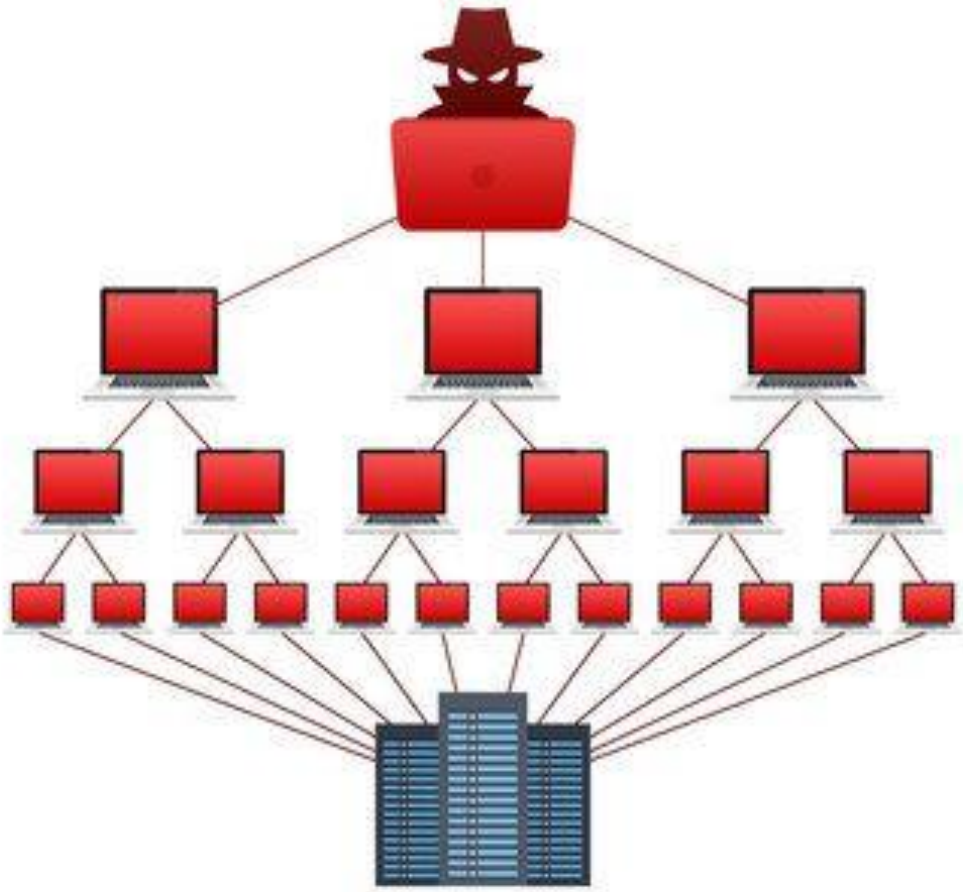
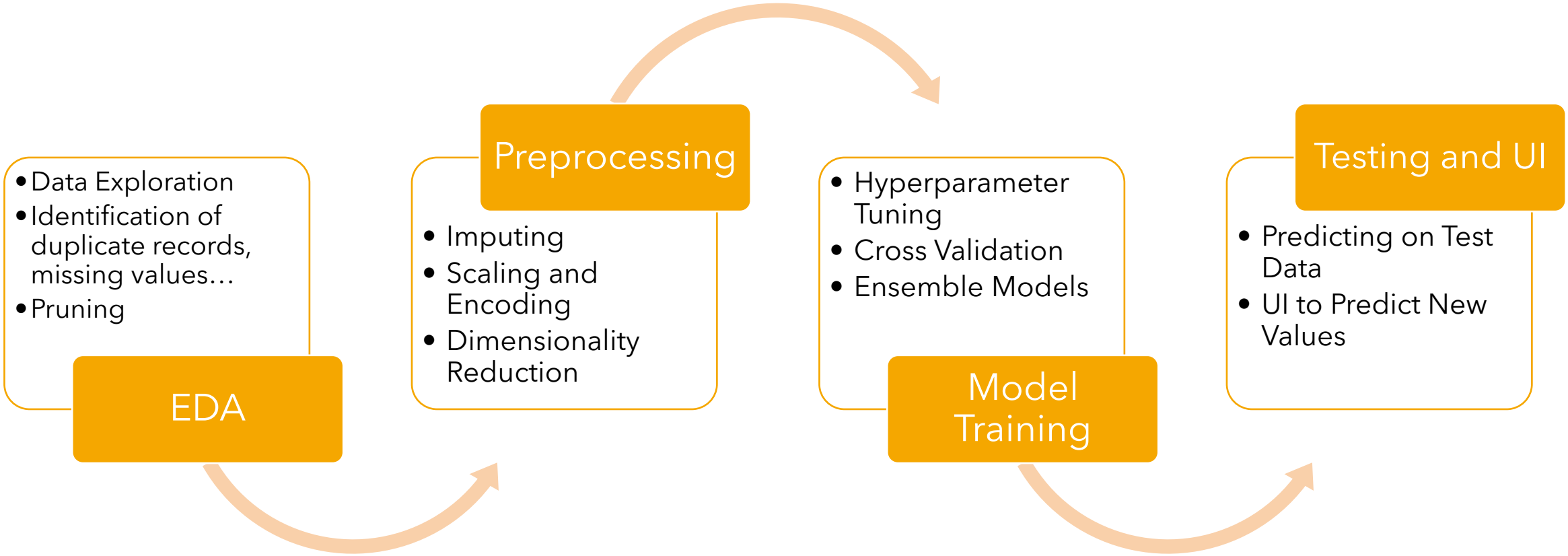
DDOS Attack Detection



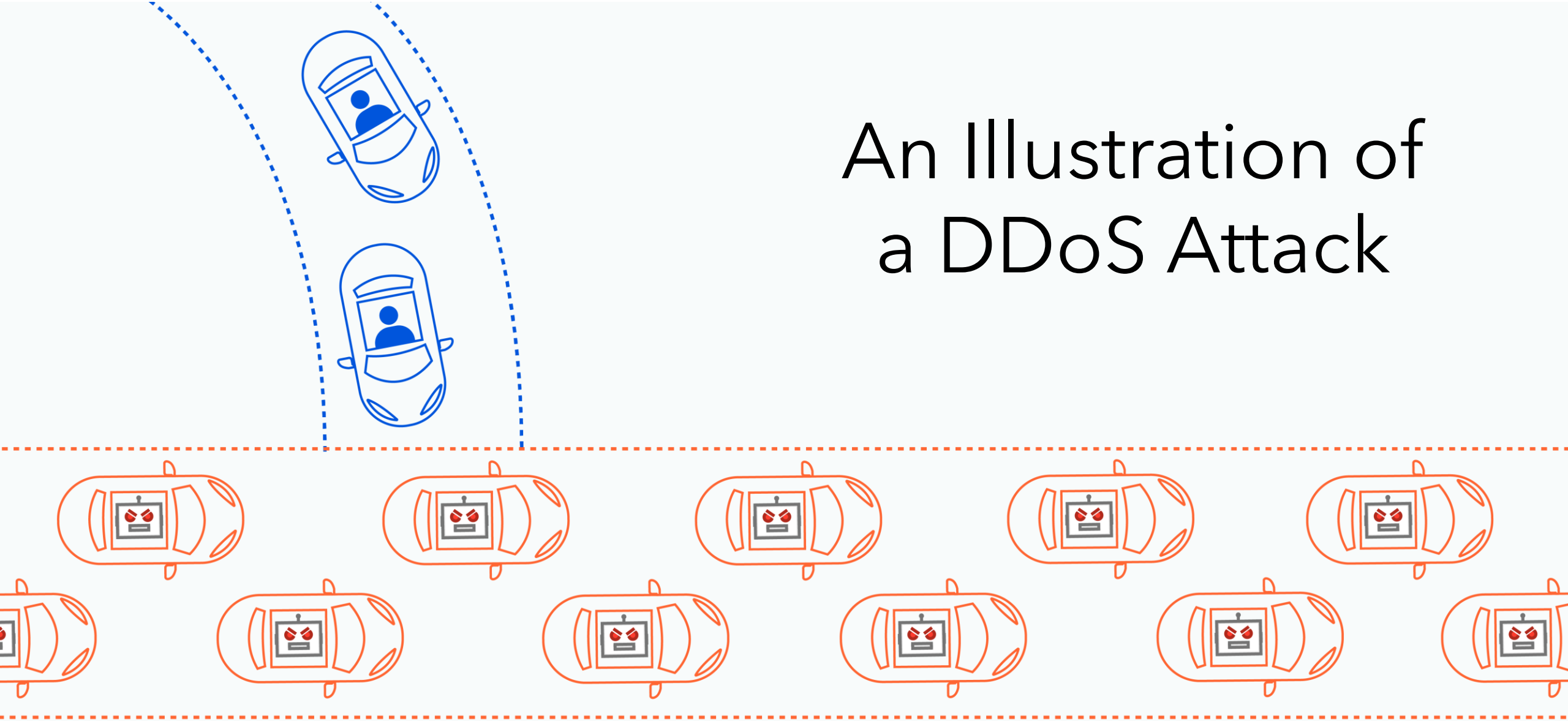
Yuvaraj K	2022506079
Nithish S	2022506099
Krishnaa S	2022506104

Flow Diagram



DoS	DDoS
Denial of Service	Distributed Denial of Service
A malicious attempt to disrupt the normal traffic of a targeted server/service/network by overwhelming the target or its surrounding infrastructure with a flood of Internet traffic	
Uses a single computer to launch the attack	A DoS attack that comes from many sources

An Illustration of a DDoS Attack



Working	Identification
Bots/Zombies: Infected Devices	Site/Service suddenly becoming slow
Group of Bots: Botnet	Suspicious amount of traffic originating from a single IP address or range
Attacker sends remote instructions to bots	A flood of traffic from users who share a single behavioral profile (device type/geolocation)
Each bot sends requests to target's IP address	Unexplained surge in requests to a single page
Each bot is a legitimate internet device	Odd traffic patterns, unnatural patterns and spikes of traffic

EDA and Pruning

Shape	(12794627, 84)
Identifiers	Flow ID, Source IP, Destination IP
Traffic and Size	Flow Duration, Total Packets
Flow Rates	Byte Rate, Packet Rate
TCP Flags	No. of Packets with FIN, SYN, RST…
Bulk Stats	Average No. of Packets, Bytes

Snippet of the Dataset

First 5 Records, First 7 and Last 7 Attributes

	Flow ID	Src IP	Src Port	Dst IP	Dst Port	Protocol	Timestamp	Active Max	Active Min	Idle Mean	Idle Std	Idle Max	Idle Min	Label
0	192.168.4.118-203.73.24.75-4504-80-6	192.168.4.118	4504	203.73.24.75	80	6	12/06/2010 08:34:32 AM	0.00	0.00	0.00	0.00	0.00	0.00	ddos
1	192.168.4.118-203.73.24.75-4504-80-6	192.168.4.118	4504	203.73.24.75	80	6	12/06/2010 08:34:36 AM	0.00	0.00	0.00	0.00	0.00	0.00	ddos
2	192.168.4.118-203.73.24.75-4505-80-6	192.168.4.118	4505	203.73.24.75	80	6	12/06/2010 08:34:36 AM	0.00	0.00	0.00	0.00	0.00	0.00	ddos
3	192.168.4.118-203.73.24.75-4505-80-6	192.168.4.118	4505	203.73.24.75	80	6	12/06/2010 08:34:37 AM	0.00	0.00	0.00	0.00	0.00	0.00	ddos
4	192.168.4.118-203.73.24.75-4506-80-6	192.168.4.118	4506	203.73.24.75	80	6	12/06/2010 08:34:37 AM	0.00	0.00	0.00	0.00	0.00	0.00	ddos

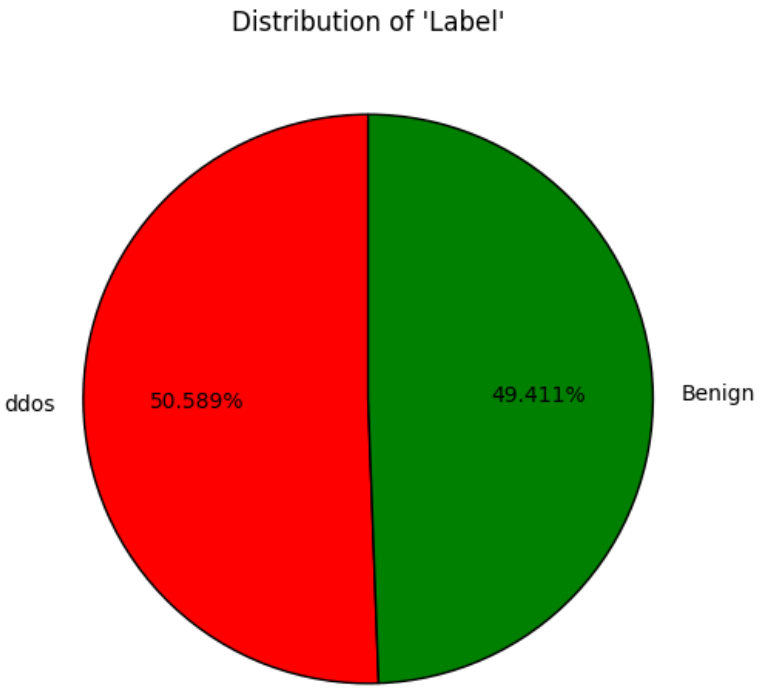
Summary Statistics of Numerical Features

	count	mean	std	min	25%	50%	75%	max
Src Port	12794627.00	37070.54	25219.85	0.00	443.00	50592.00	56215.00	65535.00
Dst Port	12794627.00	14642.90	23063.83	0.00	80.00	80.00	38550.00	65535.00
Protocol	12794627.00	7.83	4.21	0.00	6.00	6.00	6.00	17.00
Flow Duration	12794627.00	8219593.07	24773266.93	-1.00	1262.00	32065.00	4159735.50	120000000.00
Tot Fwd Pkts	12794627.00	27.20	1720.58	0.00	1.00	2.00	4.00	309628.00
...	...	...	...	...	...	...	...	...
Active Min	12794627.00	90305.39	1934856.16	0.00	0.00	0.00	0.00	113269143.00
Idle Mean	12794627.00	3119925.05	12192610.76	0.00	0.00	0.00	0.00	119999735.00
Idle Std	12794627.00	109353.20	1414692.79	0.00	0.00	0.00	0.00	76393949.17
Idle Max	12794627.00	3215789.64	12444759.34	0.00	0.00	0.00	0.00	119999735.00
Idle Min	12794627.00	3019046.02	12065274.42	0.00	0.00	0.00	0.00	119999735.00

79 rows × 8 columns

Out of 84 features, 79 are actually numerical.

Output Distribution

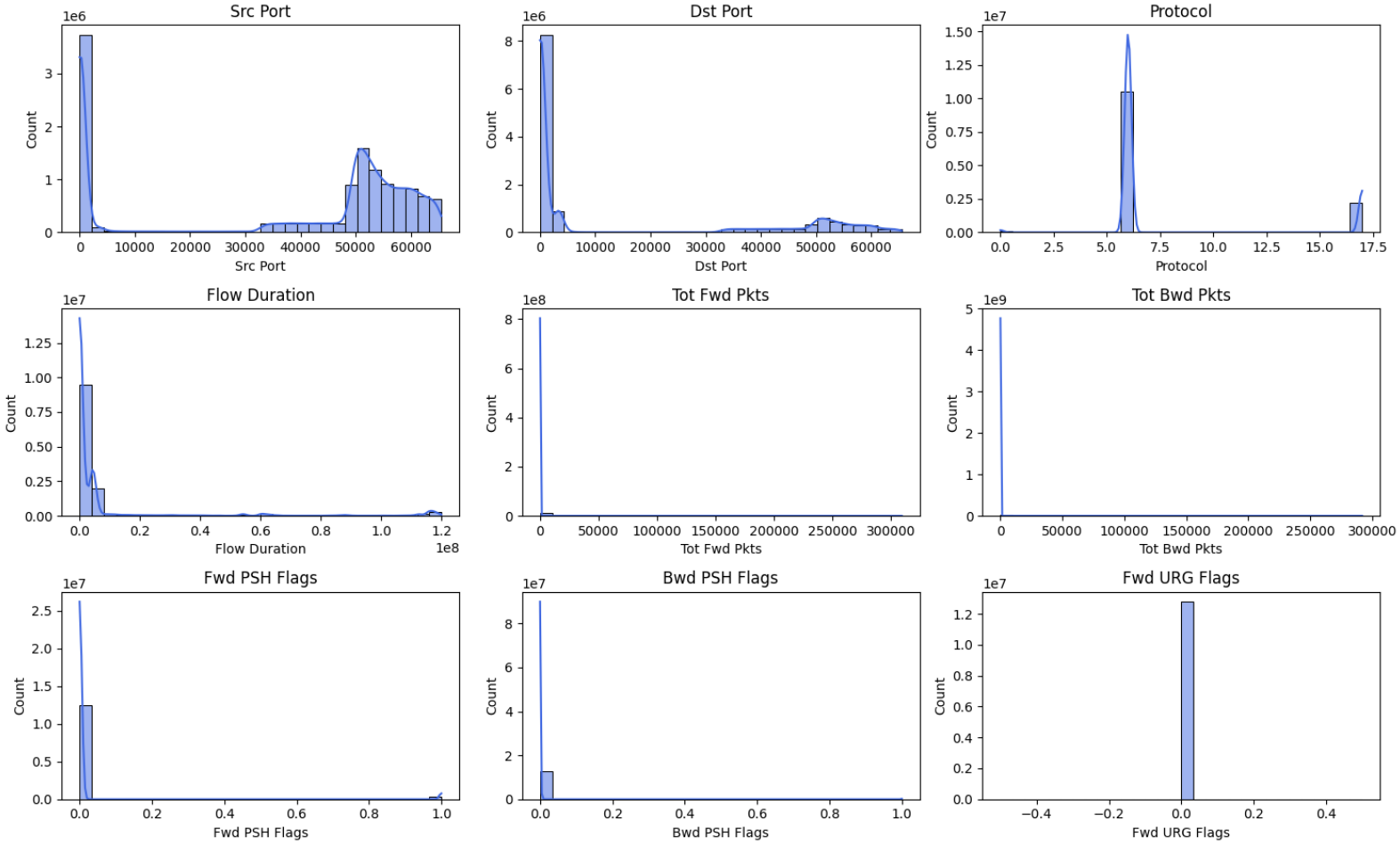


There's nearly a 50-50 split of 'ddos' and 'benign', meaning the data is balanced. There is NO class imbalance.

ddos: the given flow is a DDoS attack
benign: the given flow is regular traffic

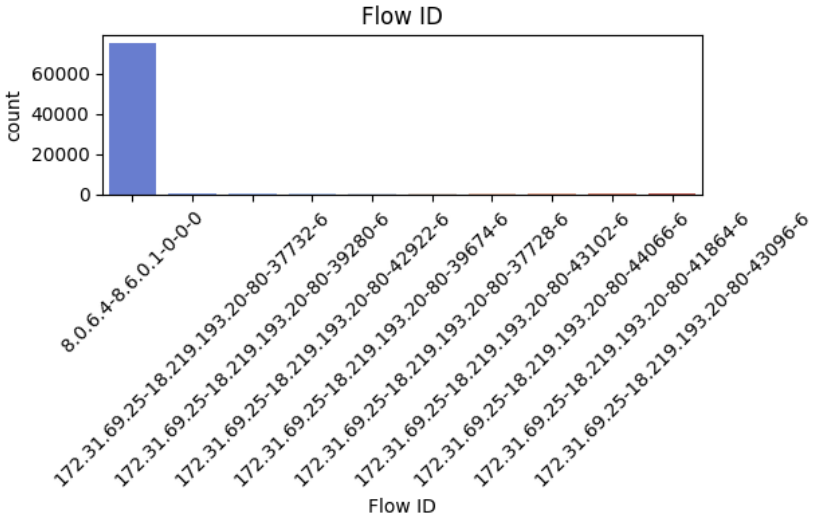
Distribution of Numerical Features

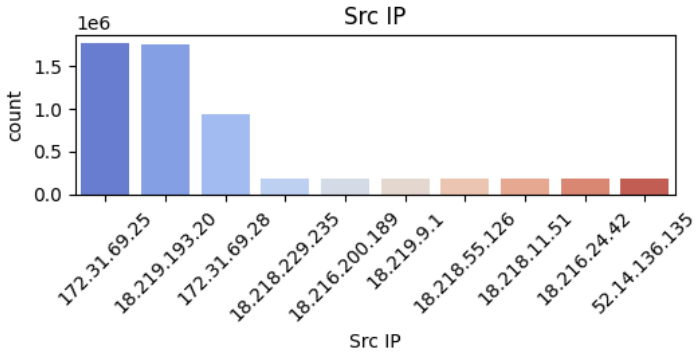
The distribution of the first 9 numerical features.

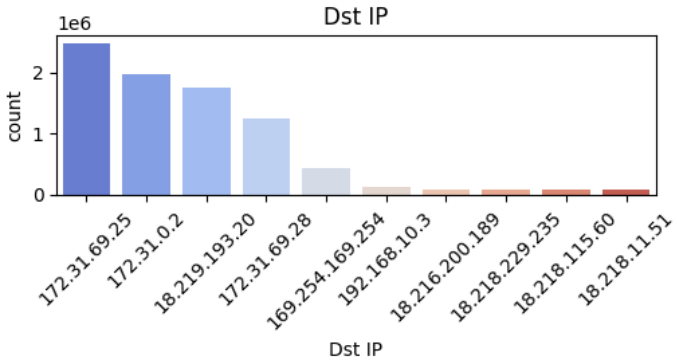


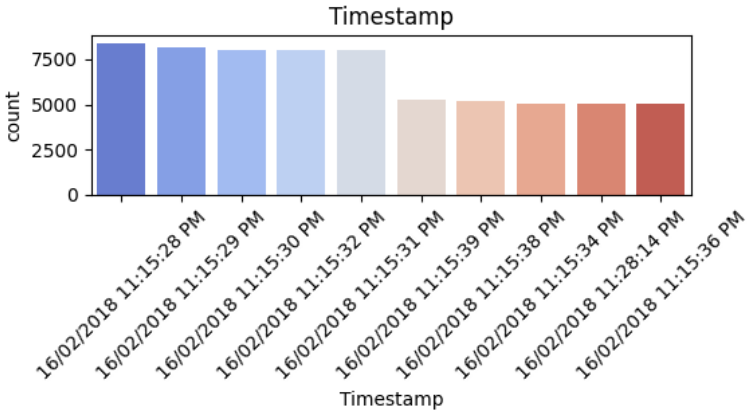
Distribution of Categorical Features

The distribution of the all the 4 categorical features (excluding, of course, the output variable ‘Label’).



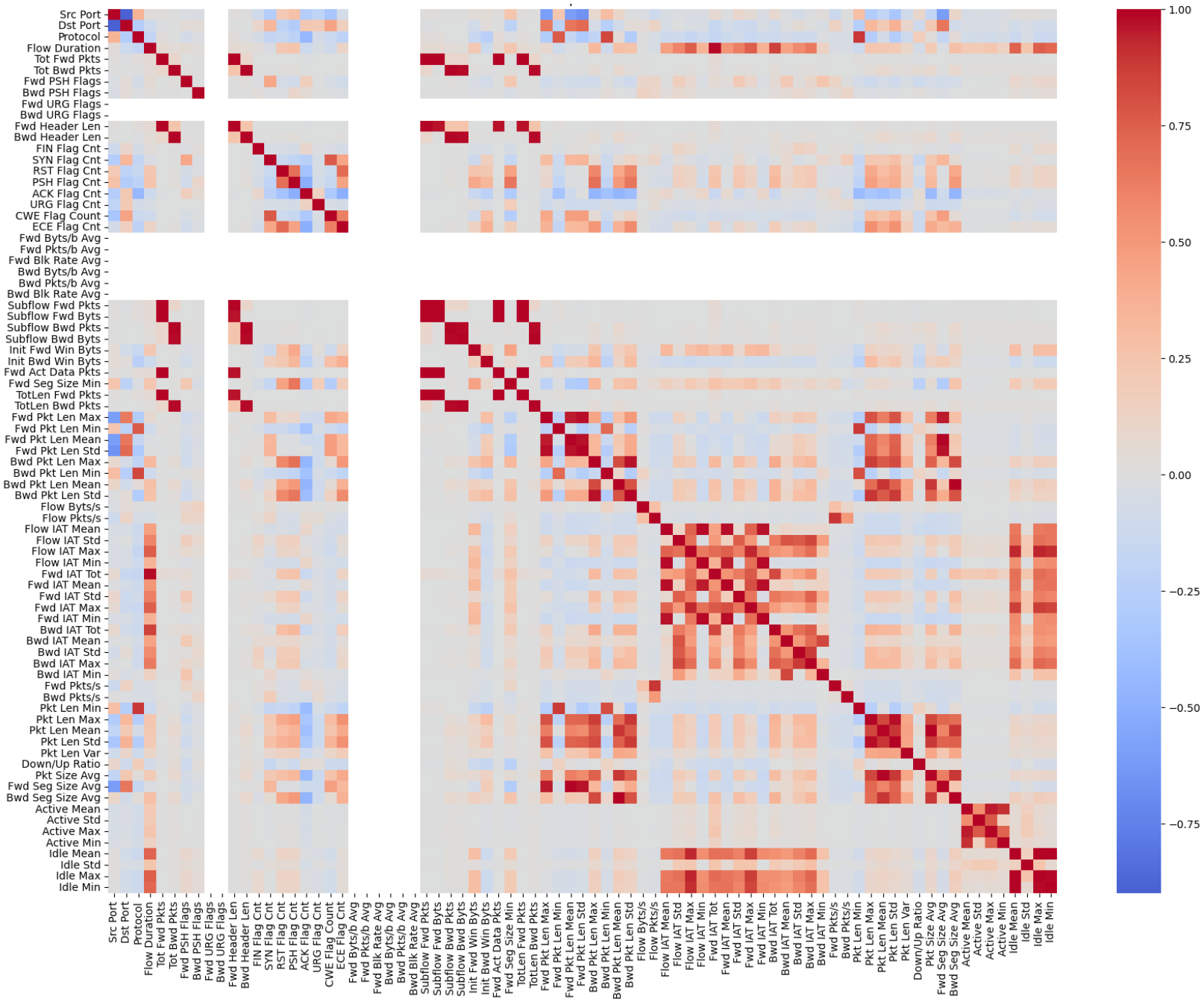






Correlation Heatmap

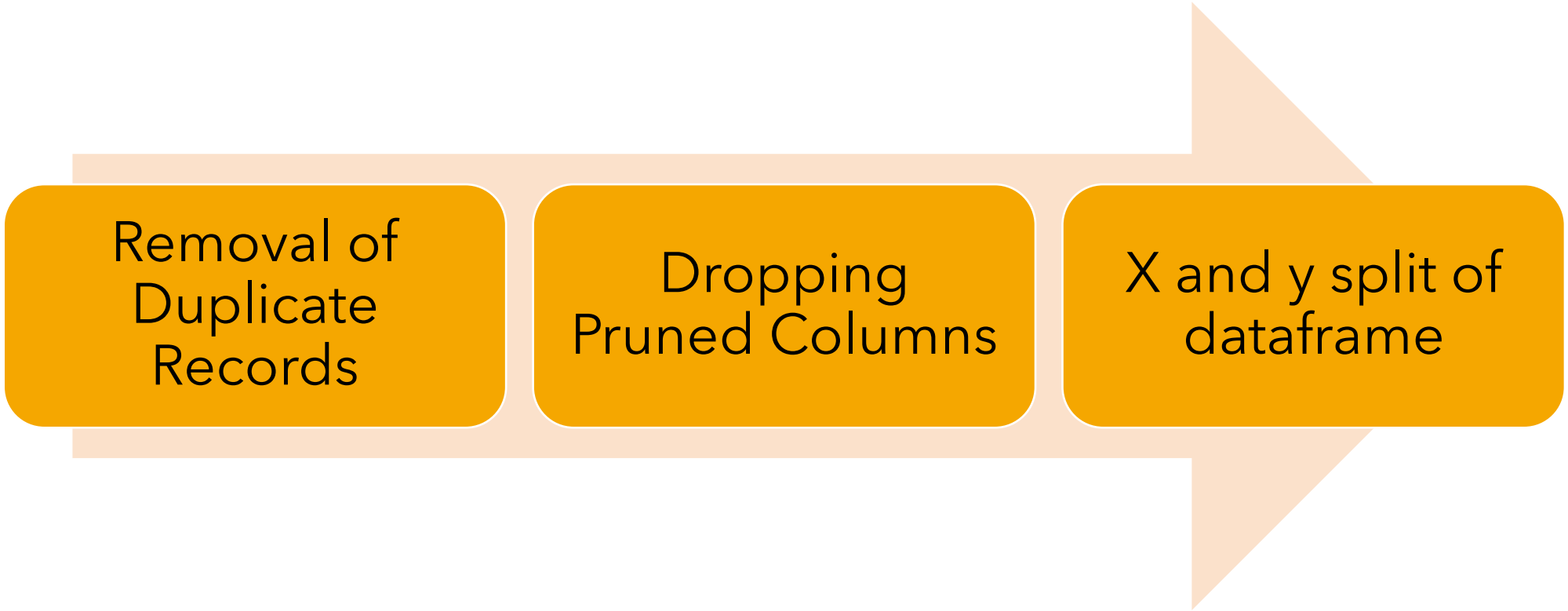
Closer to 1	Highly Correlated
Closer to 0	No Correlation
Closer to -1	Inversely Correlated



Summary of EDA

- Balanced dataset
- 3 different types of data: object, int64, float64 + (time data)
- 29713 missing values, all in the feature Flow Byts/s
- 25 duplicate records
- Good spread of categorical data
- In numerical data, single values dominate some features
- One value columns and 85% dominant columns are computed
- High +vely or –vely correlated features are pruned

Preprocessing



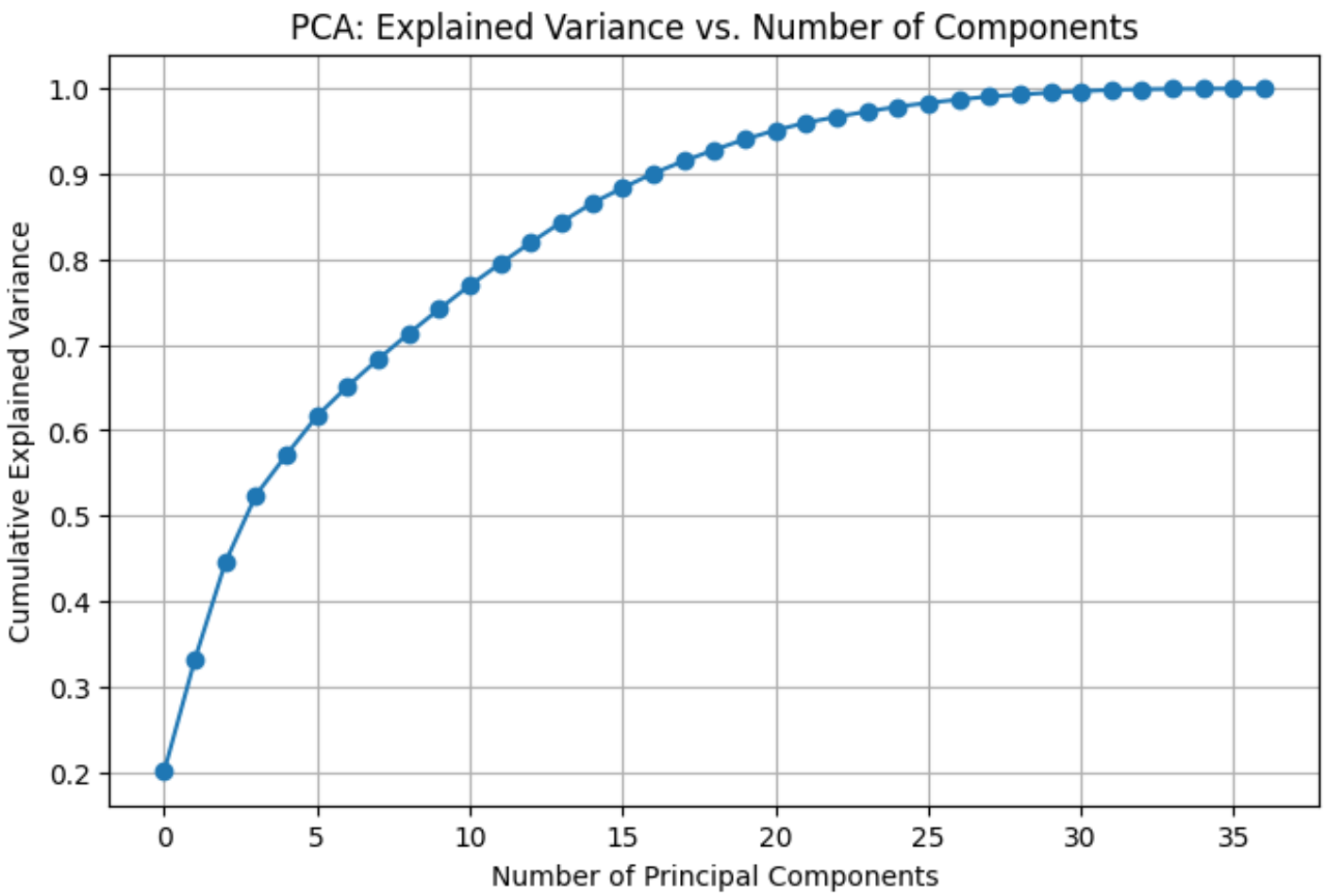
Encoding and Scaling

Label Encoding Categorical Data

Ordinal Encoding of Time Data

Standard Scaling

Is PCA Viable?



About 20 or so components seem to explain 95% of the variance in the data. So, PCA is viable!

Reduced Data

```
array([[5.07, -1.58, -5.49, ..., -2.45, 3.67, -0.04],
       [-1.51, -1.23, 1.79, ..., 0.30, -0.14, -0.31],
       [2.22, -1.48, -2.55, ..., -0.46, 0.67, -0.39],
       ...,
       [-2.32, 3.70, -1.13, ..., 0.49, -0.50, -0.13],
       [-2.31, 3.61, -1.33, ..., -0.15, 0.10, 0.14],
       [-1.60, 0.54, 0.66, ..., -0.75, -0.41, -0.45]],
      shape=(12794602, 20))
```

What Next?

- Usage of ensemble models to train data
- Hyperparameter tuning
- Cross validation
- Testing and Interface